A Hölder Counterexample to the Proposed $O(\log n/n)$ Evolution–Trotter Rate

Autonomous proof attempt for arXiv:2002.04483

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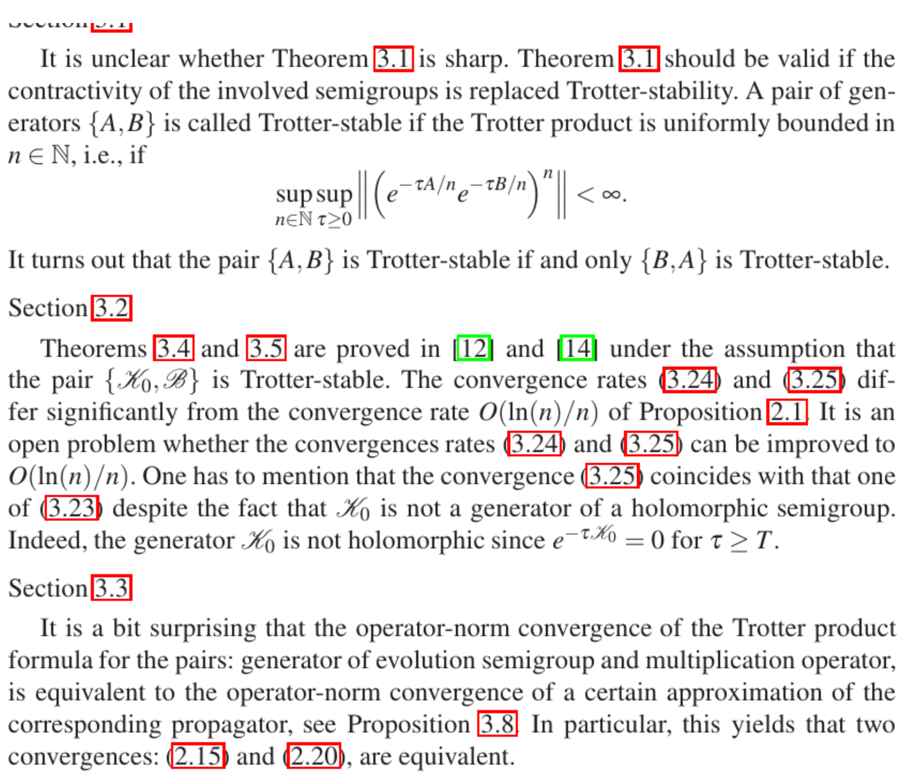
Status: counterexample likely valid, pending human review. For every $0 < \beta < 1$, a scalar commuting example satisfying assumptions (A1)–(A5) of the source has error $\Omega(n^{-\beta})$ along dyadic n. Hence the Hölder-time estimate (3.24) cannot in general be improved to $O(\log n/n)$. The separate Lipschitz question for (3.25) is not settled.

1 Source question

The source is H. Neidhardt, A. Stephan, and V. A. Zagrebnov, *Operator-Norm Convergence of the Trotter Product Formula on Hilbert and Banach Spaces: A Short Survey*, arXiv:2002.04483v1 (2020) [1]. In Section 3.4, page 14, after Theorems 3.4–3.5, the authors ask whether the rates (3.24) and (3.25) can be improved to $O(\log n/n)$.


Section 3.1

It is unclear whether Theorem 3.1 is sharp. Theorem 3.1 should be valid if the contractivity of the involved semigroups is replaced Trotter-stability. A pair of generators $\{A,B\}$ is called Trotter-stable if the Trotter product is uniformly bounded in $n \in \mathbb{N}$, i.e., if

$$\sup_{n\in\mathbb{N}} \sup_{\tau\geq 0} \left\| \left(e^{-\tau A/n} e^{-\tau B/n}\right)^n \right\| < \infty.$$

It turns out that the pair $\{A,B\}$ is Trotter-stable if and only $\{B,A\}$ is Trotter-stable.

Section 3.2

Theorems 3.4 and 3.5 are proved in [12] and [14] under the assumption that the pair $\{\mathscr{K}_0, \mathscr{B}\}$ is Trotter-stable. The convergence rates (3.24) and (3.25) differ significantly from the convergence rate $O(\ln(n)/n)$ of Proposition 2.1. It is an open problem whether the convergences rates (3.24) and (3.25) can be improved to $O(\ln(n)/n)$. One has to mention that the convergence (3.25) coincides with that one of (3.23) despite the fact that $\mathscr{K}_0$ is not a generator of a holomorphic semigroup. Indeed, the generator $\mathscr{K}_0$ is not holomorphic since $e^{-\tau\mathscr{K}_0} = 0$ for $\tau \geq T$.

Section 3.3

It is a bit surprising that the operator-norm convergence of the Trotter product formula for the pairs: generator of evolution semigroup and multiplication operator, is equivalent to the operator-norm convergence of a certain approximation of the corresponding propagator, see Proposition 3.8. In particular, this yields that two convergences: (2.15) and (2.20), are equivalent.

Figure 1: The exact open-question passage in arXiv:2002.04483v1, p. 14.

Theorem 3.4 assumes a Hölder exponent $\beta \in (\alpha, 1)$ and gives $O(n^{-(\beta-\alpha)})$. The construction below disproves the proposed $O(\log n/n)$ improvement for this Hölder branch. It works in the scalar model that the source itself identifies in Section 4.3 as a special case of (A1)–(A5).

2 Idea of the proof

For a scalar time-dependent multiplication operator, evolution–Trotter error is precisely the error of a uniform left- or right-endpoint Riemann sum. We build a nonnegative periodic Hölder function from dyadic tent waves. On a grid with $n = 2^M$ points, every wave of frequency below 2^M has exactly the correct discrete average, while every wave of frequency at least 2^M vanishes at all grid points. The whole unresolved tail is therefore missed, and its integral is exactly a positive constant times $n^{-\beta}$.

3 Counterexample theorem

Let $T = 3$, $X = \mathbb{C}$, and let D_0 be the generator of the right-shift semigroup on $L^2([0, T])$. Fix $0 < \alpha < \beta < 1$, put $A = I$, and write $\mathcal{K}_0 = D_0 + I$. Define the one-periodic tent function and its Takagi–Landsberg series by

$$\phi(x) = \operatorname{dist}(x, \mathbb{Z}), \qquad q(t) = \sum_{m=0}^{\infty} 2^{-m\beta} \phi(2^m t).$$

Let $B(t) = q(t)I$, let $\mathcal{Q}$ be multiplication by q, and let $\mathcal{K} = D_0 + I + \mathcal{Q}$.

Theorem 1. *The family $(A, B(t))$ satisfies assumptions (A1)–(A5) of [1, Assumption 3.3], with the chosen α and β. There is a constant $c_\beta > 0$ such that, for every $M \geq 1$ and $n = 2^M$,*

$$\sup_{\tau \geq 0} \left\| (e^{-\tau\mathcal{Q}/n} e^{-\tau\mathcal{K}_0/n})^n - e^{-\tau\mathcal{K}} \right\| \geq c_\beta n^{-\beta},$$
$$\sup_{\tau \geq 0} \left\| (e^{-\tau\mathcal{K}_0/n} e^{-\tau\mathcal{Q}/n})^n - e^{-\tau\mathcal{K}} \right\| \geq c_\beta n^{-\beta}.$$

In particular, neither error is $O(\log n/n)$.

Proof. We divide the proof into four short steps.

Step 1: q is nonnegative, bounded, and β-Hölder. The first two assertions follow from $0 \leq \phi \leq 1/2$ and uniform convergence. Let $h = |t - s| \in (0, 1]$ and choose $J \geq 0$ so that $2^J h \leq 1 < 2^{J+1} h$. Since ϕ is 1-Lipschitz,

$$\begin{aligned} |q(t) - q(s)| &\leq h \sum_{m=0}^{J} 2^{m(1-\beta)} + \sum_{m>J} 2^{-m\beta} \\ &\leq C_\beta h^\beta. \end{aligned}$$

For $h > 1$ the same estimate follows after enlarging C_β. Thus $q \in C^{0,\beta}([0, 3])$.

Step 2: all source hypotheses hold. The scalar $A = I$ is invertible and generates a holomorphic contraction semigroup. Each $B(t) = q(t)I$ is bounded, nonnegative, and generates the contraction $e^{-\tau q(t)}I$. All domain, measurability, and adjoint requirements in (A2)–(A4) are automatic. Since every power of A is I, (A5) is exactly

$$\|A^{-1}(B(t) - B(s))A^{-\alpha}\| = |q(t) - q(s)| \leq C_\beta |t - s|^\beta.$$

The pair is also Trotter-stable because all factors are contractions.

Step 3: exact dyadic quadrature defect. Over any interval of length one,

$$Q_\beta := \int_1^2 q(t)\,dt = \frac{1}{4}\sum_{m=0}^{\infty} 2^{-m\beta} = \frac{1}{4(1-2^{-\beta})},$$

because $\int_0^1 \phi(u)\,du = 1/4$. Take $n = 2^M$. For $m < M$, the grid average of $\phi(2^m t)$ on $[1,2]$ is $1/4$: after removing repeated periods it is

$$\frac{1}{L}\sum_{j=0}^{L-1} \phi(j/L) = \frac{1}{4}, \qquad L = 2^{M-m} \geq 2.$$

For $m \geq M$, every grid value is zero. Hence both endpoint sums agree and

$$\begin{aligned} R_M &:= \frac{1}{n}\sum_{k=0}^{n-1} q(1+k/n) = \frac{1}{n}\sum_{k=1}^{n} q(1+k/n) \\ &= \frac{1}{4}\sum_{m=0}^{M-1} 2^{-m\beta}, \end{aligned}$$

so that

$$\delta_M := Q_\beta - R_M = \frac{n^{-\beta}}{4(1-2^{-\beta})}. \tag{1}$$

Step 4: transfer the defect to operator norm. The exact propagator for $D_0 + I + \mathcal{Q}$ on the interval $[s,t]$ is the scalar multiplier

$$U(t,s) = \exp\left(-(t-s) - \int_s^t q(y)\,dy\right).$$

The two product orders replace the integral by the left- and right-endpoint sums, respectively. At the interior point $(t,s) = (2,1)$ those sums both equal R_M. Proposition 3.8 of the source identifies the evolution-semigroup operator norm with the essential supremum of the propagator error. The multipliers are continuous, so this essential supremum is at least their value at $(2,1)$. By (1), for either product order it is at least

$$\begin{aligned} e^{-1-R_M} - e^{-1-Q_\beta} &= e^{-1-Q_\beta}(e^{\delta_M} - 1) \\ &\geq \frac{e^{-1-Q_\beta}}{4(1-2^{-\beta})}\, n^{-\beta}. \end{aligned}$$

This proves the asserted lower bound. Finally, $n^{-\beta}/(\log n/n) = n^{1-\beta}/\log n \to \infty$, so an $O(\log n/n)$ estimate is impossible. □

4 Verification and scope

The script `code/check_dyadic_identity.py` checks finite truncations of the exact quadrature identity, the constancy of $n^\beta \delta_M$, and the semigroup lower bound. It is verification only; the proof above is exact.

The result gives a complete negative answer to the (A5), $0 < \beta < 1$ half of the source's proposed $O(\log n/n)$ improvement. It does not address the separate Lipschitz assumption (A6) or whether the general upper rate $O(n^{-(\beta-\alpha)})$ can be sharpened to the scalar-optimal $O(n^{-\beta})$.

5 Bounded novelty check

We searched the run indexes and exact arXiv id; the full sources of arXiv:1612.06147, 1703.09536, 1901.02205, and 2002.04483; and arXiv API queries through 11 August 2026 for combinations of "Trotter product", "Hölder", "non-autonomous/evolution", "sharp", and "Riemann sums". The cited scalar paper proves the upper $O(n^{-\beta})$ estimate and arbitrary slowness for merely continuous functions, but no Hölder lower example or application disproving the survey's open $O(\log n/n)$ question was found. The novelty claim is necessarily bounded; the dyadic construction is an elementary sharp quadrature example and may be known outside the searched Trotter literature.

Human-review recommendation. Check the source's quantifier in the plural open question and the transfer from the interior propagator value to the essential supremum. If confirmed, accept this as a counterexample to the Hölder branch, explicitly retaining the A6 limitation.

References

[1] H. Neidhardt, A. Stephan, and V. A. Zagrebnov, *Operator-Norm Convergence of the Trotter Product Formula on Hilbert and Banach Spaces: A Short Survey*, arXiv:2002.04483v1, 2020.

[2] H. Neidhardt, A. Stephan, and V. A. Zagrebnov, *Remarks on the operator-norm convergence of the Trotter product formula*, arXiv:1703.09536v1, 2017.